

1 覚えるべき三角形の図示 解答

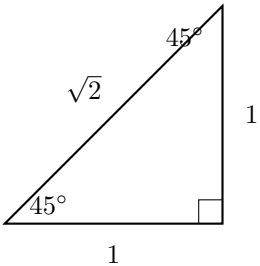
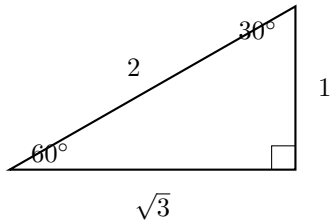
1. 30° - 60° - 90° の辺比 $1 : \sqrt{3} : 2$
2. 45° - 45° - 90° の辺比 $1 : 1 : \sqrt{2}$
3. 最短辺 $= 1 \Rightarrow (1, \sqrt{3}, 2)$
4. 斜辺 $= 1 \Rightarrow \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 1\right)$
5. $\sin 30^{\circ} = \frac{1}{2}, \cos 30^{\circ} = \frac{\sqrt{3}}{2}, \sin 45^{\circ} = \cos 45^{\circ} = \frac{\sqrt{2}}{2}, \sin 60^{\circ} = \frac{\sqrt{3}}{2}, \cos 60^{\circ} = \frac{1}{2}$
6. $\tan 30^{\circ} = \frac{1}{\sqrt{3}}, \tan 45^{\circ} = 1, \tan 60^{\circ} = \sqrt{3}$

考え方 基準三角形の辺比を先に固定する。 30° - 60° - 90° は $1 : \sqrt{3} : 2$ 、 45° - 45° - 90° は $1 : 1 : \sqrt{2}$ 。

途中式（代表例）

$$\sin 30^{\circ} = \frac{\text{対辺}}{\text{斜辺}} = \frac{1}{2}, \cos 30^{\circ} = \frac{\sqrt{3}}{2}$$
$$\tan 60^{\circ} = \frac{\text{対辺}}{\text{隣辺}} = \frac{\sqrt{3}}{1} = \sqrt{3}$$

図説（覚えるべき 2 つの三角形）



2 直角三角形の比 解答

1. $2, 2\sqrt{3}, 4$
2. $4, 4\sqrt{3}, 8$
3. $6, 6, 6\sqrt{2}$
4. 30° 側 $= 3$, 斜辺 $= 6$
5. 60° 側 $= 5\sqrt{3}$, 斜辺 $= 10$
6. もう 1 辺 $= 4$, 斜辺 $= 4\sqrt{2}$

考え方 与えられた辺を基準比に対応させ、倍率をかける。

途中式（代表例）

1. 30 - 60 - 90 比 $1 : \sqrt{3} : 2$ で最短辺 $2 \Rightarrow \times 2$
 $(2, 2\sqrt{3}, 4)$
3. 45 - 45 - 90 比 $1 : 1 : \sqrt{2}$, 斜辺 $6\sqrt{2} \Rightarrow$ 倍率 6
 $(6, 6, 6\sqrt{2})$

3 三角比 解答

1. $\cos 30^\circ = \frac{\sqrt{3}}{2}, \cos 45^\circ = \frac{\sqrt{2}}{2}, \cos 60^\circ = \frac{1}{2}$

2. $\sin 30^\circ = \frac{1}{2}, \sin 45^\circ = \frac{\sqrt{2}}{2}, \sin 60^\circ = \frac{\sqrt{3}}{2}$

3. $\tan 30^\circ = \frac{1}{\sqrt{3}}, \tan 45^\circ = 1, \tan 60^\circ = \sqrt{3}$

4. 底辺 = $4\sqrt{3}$, 高さ = 4

5. 底辺 = 5, 高さ = $5\sqrt{3}$

6. 斜辺 = $6\sqrt{2}$, 高さ = 6

考え方 $\sin \theta = \frac{\text{高さ}}{\text{斜辺}}, \cos \theta = \frac{\text{底辺}}{\text{斜辺}}, \tan \theta = \frac{\text{高さ}}{\text{底辺}}$ を使う。

途中式（代表例）

4. $\theta = 30^\circ$, 斜辺 8 $\Rightarrow$ 高さ = $8 \sin 30^\circ = 8 \cdot \frac{1}{2} = 4$

底辺 = $8 \cos 30^\circ = 8 \cdot \frac{\sqrt{3}}{2} = 4\sqrt{3}$

6. $\theta = 45^\circ$, $\tan 45^\circ = 1 \Rightarrow$ 高さ = 底辺 = 6

斜辺 = $\sqrt{6^2 + 6^2} = 6\sqrt{2}$

4 ラジアン 解答

1. $30^\circ = \frac{\pi}{6}, 45^\circ = \frac{\pi}{4}, 60^\circ = \frac{\pi}{3}, 90^\circ = \frac{\pi}{2}$

2. $\frac{\pi}{6} = 30^\circ, \frac{\pi}{4} = 45^\circ, \frac{\pi}{3} = 60^\circ, \frac{\pi}{2} = 90^\circ$

3. $\cos \frac{\pi}{3} = \frac{1}{2}, \sin \frac{\pi}{6} = \frac{1}{2}, \tan \frac{\pi}{4} = 1$

4. 底辺 = $4\sqrt{3}$, 高さ = 4

5. 底辺 = $2\sqrt{2}$, 高さ = $2\sqrt{2}$

6. 斜辺 = 6, 高さ = $3\sqrt{3}$

考え方 $180^\circ = \pi$ を基準に変換。辺計算は度数法と同じで角度表記だけが違う。

途中式（代表例）

1. $60^\circ = \frac{60}{180}\pi = \frac{\pi}{3}$

2. $\frac{\pi}{4} = \frac{180^\circ}{4} = 45^\circ$

6. $\theta = \frac{\pi}{3} (= 60^\circ), \cos \theta = \frac{1}{2} = \frac{\text{底辺}}{\text{斜辺}} = \frac{3}{\text{斜辺}}$

斜辺 = 6, 高さ = $6 \sin \frac{\pi}{3} = 6 \cdot \frac{\sqrt{3}}{2} = 3\sqrt{3}$

5 単位円の三角比 解答

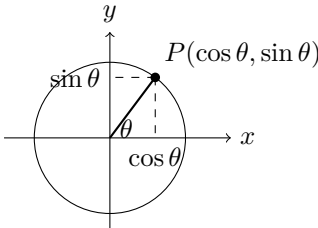
1. $\sin \theta = \frac{\sqrt{3}}{2}, \cos \theta = \frac{1}{2}, \tan \theta = \sqrt{3}$
2. $\sin \theta = \frac{\sqrt{2}}{2}, \cos \theta = \frac{\sqrt{2}}{2}, \tan \theta = 1$
3. $\sin \theta = \frac{\sqrt{3}}{2}, \cos \theta = -\frac{1}{2}, \tan \theta = -\sqrt{3}$
4. $\cos \theta = \pm \frac{2\sqrt{2}}{3}, \tan \theta = \pm \frac{\sqrt{2}}{4}$
5. $\sin \theta = \frac{\sqrt{15}}{4}, \tan \theta = \sqrt{15}$
6. $\sin \theta = \frac{2\sqrt{5}}{5}, \cos \theta = \frac{\sqrt{5}}{5}$

考え方 単位円では $\cos \theta = x, \sin \theta = y, \tan \theta = \frac{y}{x}$ 。また $\sin^2 \theta + \cos^2 \theta = 1$ を使う。

途中式（代表例）

3. $P\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \Rightarrow \cos \theta = -\frac{1}{2}, \sin \theta = \frac{\sqrt{3}}{2}$
 $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\sqrt{3}/2}{-1/2} = -\sqrt{3}$
6. $\tan \theta = 2 \Rightarrow \sin \theta = 2 \cos \theta$
 $(2 \cos \theta)^2 + \cos^2 \theta = 1 \Rightarrow 5 \cos^2 \theta = 1$
 $\cos \theta = \frac{\sqrt{5}}{5}, \sin \theta = \frac{2\sqrt{5}}{5}$

図説（単位円と三角比）



6 三角比の総合 解答

1. $\frac{1 + \sqrt{3}}{2}$
2. $\frac{1}{2}$
3. $\frac{\sqrt{3}}{2}$
4. $\sqrt{3} + 2$
5. 1
6. $\cos \theta = \frac{4}{5}, \tan \theta = \frac{3}{4}$

考え方 既知角は値を直接代入。未知角は恒等式と象限条件で符号を決める。

途中式（代表例）

4. $2 \cos \frac{\pi}{6} + 4 \sin \frac{\pi}{6} = 2 \cdot \frac{\sqrt{3}}{2} + 4 \cdot \frac{1}{2} = \sqrt{3} + 2$
6. $\sin \theta = \frac{3}{5}, 0 < \theta < \frac{\pi}{2}$
 $\cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$
 $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{3/5}{4/5} = \frac{3}{4}$